Day 44 — HR Interview AI Architecture

1. Objective

Document the existing HR Interview AI components, their responsibilities, data flow, scoring configuration, and integration boundaries in the Zecpath AI System.

2. Architecture Overview

The interview workflow is implemented through Python modules. The ATS REST API is a separate FastAPI application in `ats_engine/api.py`.

```
Candidate Profile / Role
        |
        v
InterviewAI
(interview_ai/interview.py)
        |
        +--> HR screening dataset
        |
        +--> Interview questions and phases
        |
        v
Candidate Responses
        |
        v
DynamicFollowUpEngine
(interview_ai/follow_up_engine.py)
        |
        v
HRInterviewScoringEngine
(scoring/hr_interview_scoring_engine.py)
        |
        +--> CommunicationScoringEngine
        |
        +--> ConfidenceSentimentEngine
        |
        v
Interview Evaluation
        |
        v
Interview Summary Generator
(scoring/interview_summary_generator.py)
        |
        v
Structured Recruiter Report
```

The diagram describes the documented component flow. Exact runtime orchestration should be verified against the relevant implementation before introducing a new integration.

3. Component Responsibilities

InterviewAI

Location: `interview_ai/interview.py`

Responsibilities include:

- Initializing an interview for a supported role.
- Validating experience level (`fresher` or `experienced`).
- Loading role-based questions from `data/hr_screening_dataset.json`.
- Classifying configured roles as technical or non-technical.
- Managing interview phases and state.
- Capturing responses and tracking follow-up eligibility.

The documented phases are `introduction`, `core_hr_questions`, `role_based_evaluation`, and `closing`.

DynamicFollowUpEngine

Location: `interview_ai/follow_up_engine.py`

Responsibilities include:

- Detecting empty, vague, very short, short, and detailed responses.
- Selecting clarification, deepening, example, or scenario follow-up behavior.
- Tracking follow-up state.
- Limiting follow-ups to two.
- Suppressing repeated follow-up prompts.

Follow-up decisions use configured text and response-length signals. They should not be interpreted as validated measures of a candidate's personality or ability.

HRInterviewScoringEngine

Location: `scoring/hr_interview_scoring_engine.py`

The engine combines:

- Answer relevance.
- Communication quality.
- Confidence indicators.
- Answer consistency.

Configuration: `data/hr_interview_scoring_configuration.json`.

Dimension	Weight
Answer relevance	35%

Dimension	Weight
Communication	25%
Confidence	25%
Consistency	15%

Scores are normalized to the 0?100 range. The first answer receives a consistency score of 100 when no previous answer is available; this is a default, not evidence of demonstrated consistency.

Supporting Scoring Components

- `scoring/communication_scoring_engine.py`: communication-related evaluation.
- `scoring/confidence_sentiment_engine.py`: confidence and sentiment signals.
- `scoring/aptitude_logic_engine.py`: aptitude and logic evaluation in the broader workflow.
- `scoring/interview_summary_generator.py`: creates structured and recruiter-oriented summaries from evaluation inputs.

ATS REST API

Location: `ats_engine/api.py`

The FastAPI application exposes resume upload, resume parsing, ATS scoring, shortlisting, job-status, and health endpoints. Its documented purpose is ATS processing; the existence of these routes does not establish dedicated HTTP endpoints for every HR interview operation.

4. Data Flow

1. Select a supported role and experience level.
2. Load role-specific questions from the HR screening dataset.
3. Manage question phases and capture candidate answers.
4. Analyze responses for follow-up eligibility.
5. Evaluate answer relevance, communication, confidence, and consistency.
6. Pass evaluation outputs to summary/reporting components where integrated.
7. Present results for recruiter review.

5. Configuration and Data Files

File	Purpose
`data/hr_screening_dataset.json`	Role-based interview questions
`data/hr_interview_scoring_configuration.json`	Score weights, thresholds, and range
`data/interview_summary_configuration.json`	Summary generation configuration
`data/interview_summary_template.json`	Summary report structure
`data/sample_hr_interview_summary_reports.json`	Sample generated summaries

6. API Boundary

The current REST API is implemented in `ats_engine/api.py`. The interview and scoring modules listed above are Python components. Do not describe a method as an HTTP endpoint unless a corresponding route is implemented and tested.

A future interview API could wrap interview initialization, question retrieval, response submission, and report retrieval, but such endpoints are proposals and are not documented as existing functionality here.

7. Reliability and Human Oversight

The Day 40 simulation reported 25% classification accuracy across four predefined candidate profiles. This small simulation is not a production accuracy benchmark, but it demonstrates that calibration gaps remain.

- Do not use scores as the sole basis for hiring.
- Retain human recruiter review.
- Validate score behavior using larger, diverse, recruiter-reviewed datasets.
- Treat keyword, confidence, and language signals as limited indicators.
- Protect candidate data and apply approved retention policies.

8. Related Tests

- `tests/test_day33_interview_engine.py`
- `tests/test_day37_hr_interview_scoring_engine.py`
- `tests/test_day39_interview_summary_generator.py`
- `tests/test_day40_hr_interview_simulation.py`
- `tests/test_ats_api.py`